

Punctuation Sensitivity Probe

Descriptive acoustic analysis of punctuation-prosody effects across three TTS models

— Chatterbox, XTTS-v2, and Kokoro —

over 28 utterances spanning 5 punctuation categories.

This is a preliminary smoke test, not a validated benchmark.

1 Caveat

This report describes acoustic measurements from 28 hand-crafted utterances per model. It is not a benchmark. The sample size is too small for ranking claims. All metrics are acoustic proxies (VAD pauses, pyin F0) without perceptual validation. Confidence intervals and effect sizes are reported where possible; most comparisons have 2-6 data points per condition and should be treated as anecdotal.

2 Methodology

Stimuli. 28 utterances were manually designed to probe each punctuation category. Each utterance was rendered with the exact punctuation required (no textual variation beyond the target mark) to isolate the effect of punctuation on prosody.

Categories (5). Sentence-end (period, question mark, exclamation mark), pause hierarchy (comma, semicolon, em-dash, ellipsis), trailing punctuation (ellipsis vs period), quotation (quoted speech vs reported speech), and capitalisation (ALL CAPS, Title Case, lowercase).

Models (3). Chatterbox, XTTS-v2, and Kokoro were each used to generate 28 audio clips at 24 kHz. Chatterbox and XTTS-v2 use voice cloning from VCTK p229; Kokoro uses its default voice (no adaptation baseline).

Acoustic analysis. Voice activity detection uses energy-based VAD (threshold = $0.05 \times \text{max RMS}$, 20 ms window). F0 extracted with librosa.pyin (50-600 Hz, 256-hop). Pauses are RMS-below-threshold regions ≥ 30 ms. Internal pauses are those occurring before 95% of total duration. Terminal F0 slope computed over the final 400 ms via linear regression. 95% CIs are bootstrap percentile intervals (1000 resamples).

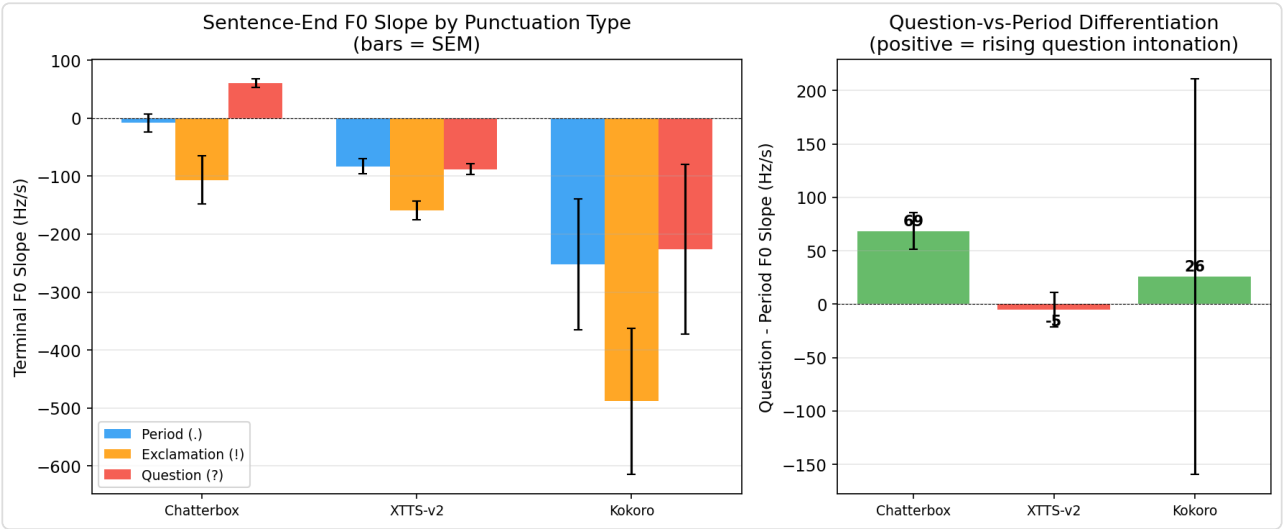
3 Gate Check

Minimum viability: period-ending utterances must have best pause ≥ 150 ms for $\geq 80\%$ of items. This confirms audible speech at boundaries, not prosodic quality.

Model	Period Items	Passing	Pass Rate	Status
Chatterbox	5	5	100%	PASS
Kokoro (no-adaptation baseline)	5	5	100%	PASS
XTTS-v2	5	5	100%	PASS

4 Sentence-End Punctuation

Mean terminal F0 slope for each punctuation type. Positive = rising (question-like), negative = falling (statement-like).



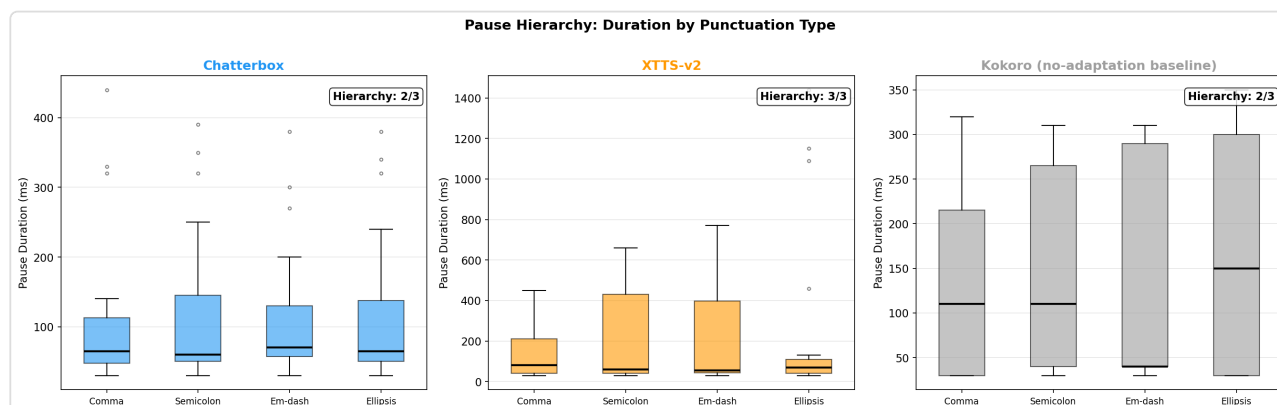
Model	Period (Hz/s)	Question (Hz/s)	Exclamation (Hz/s)	Q-P Diff (Hz/s)	Cohen's d
Chatterbox	-8.3	60.2	-106.5	68.5	2.6 (large)
XTTS-v2	-83.0	-88.0	-158.9	-5.0	-0.2 (small)
Kokoro (no-adaptation baseline)	-251.9	-226.2	-488.5	25.7	0.1 (negligible)

Chatterbox shows the largest question-mark differentiation (+68.5 Hz/s vs periods, Cohen's d large). Kokoro shows moderate differentiation (+25.7 Hz/s). XTTS-v2 is near-zero (-5.0 Hz/s, d negligible), treating

questions and statements with similar F0 trajectories. These differences are descriptive; with n=3 per condition, no significance claims are warranted.

5 Pause Hierarchy

Pause duration by internal punctuation type. Expected ordering: comma < semicolon < em-dash < ellipsis. Hierarchy score = fraction of adjacent pairs in correct order (0-1).

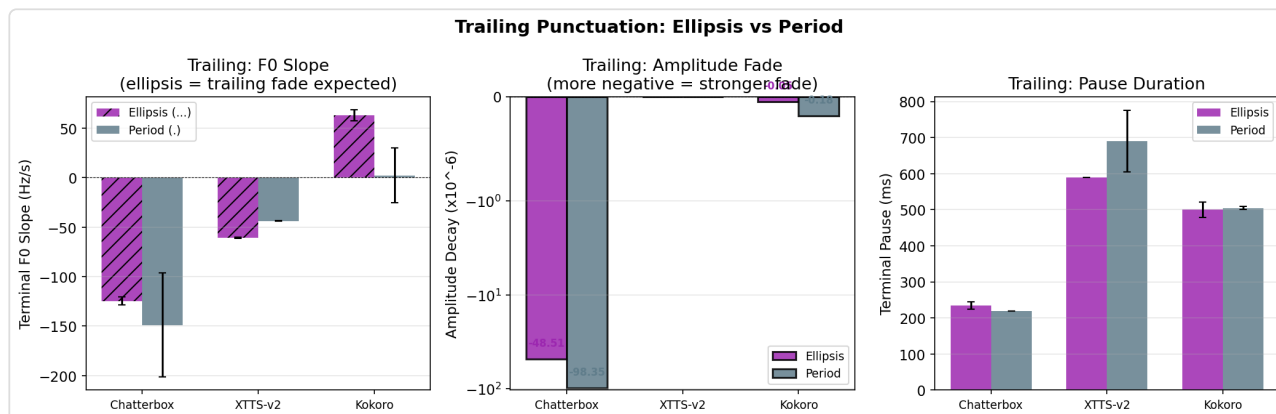


Model	Comma (ms)	Semicolon (ms)	Em-dash (ms)	Ellipsis (ms)	Hierarchy Score
Chatterbox	112.5	119.1	110.8	118.1	0.67
XTTS-v2	146.2	194.6	220.0	247.6	1.0
Kokoro (no-adaptation baseline)	135.8	154.5	126.9	166.7	0.67

XTTS-v2 achieves the highest hierarchy score (1.0) with a monotonic increase from commas to ellipses. Chatterbox and Kokoro tie at 0.67; both show inversions (em-dash is shorter than comma). With IRR at this sample size, these scores should be interpreted cautiously.

6 Trailing Punctuation: Ellipsis vs Period

F0 slope, amplitude decay, and pause duration for trailing ellipsis vs period.

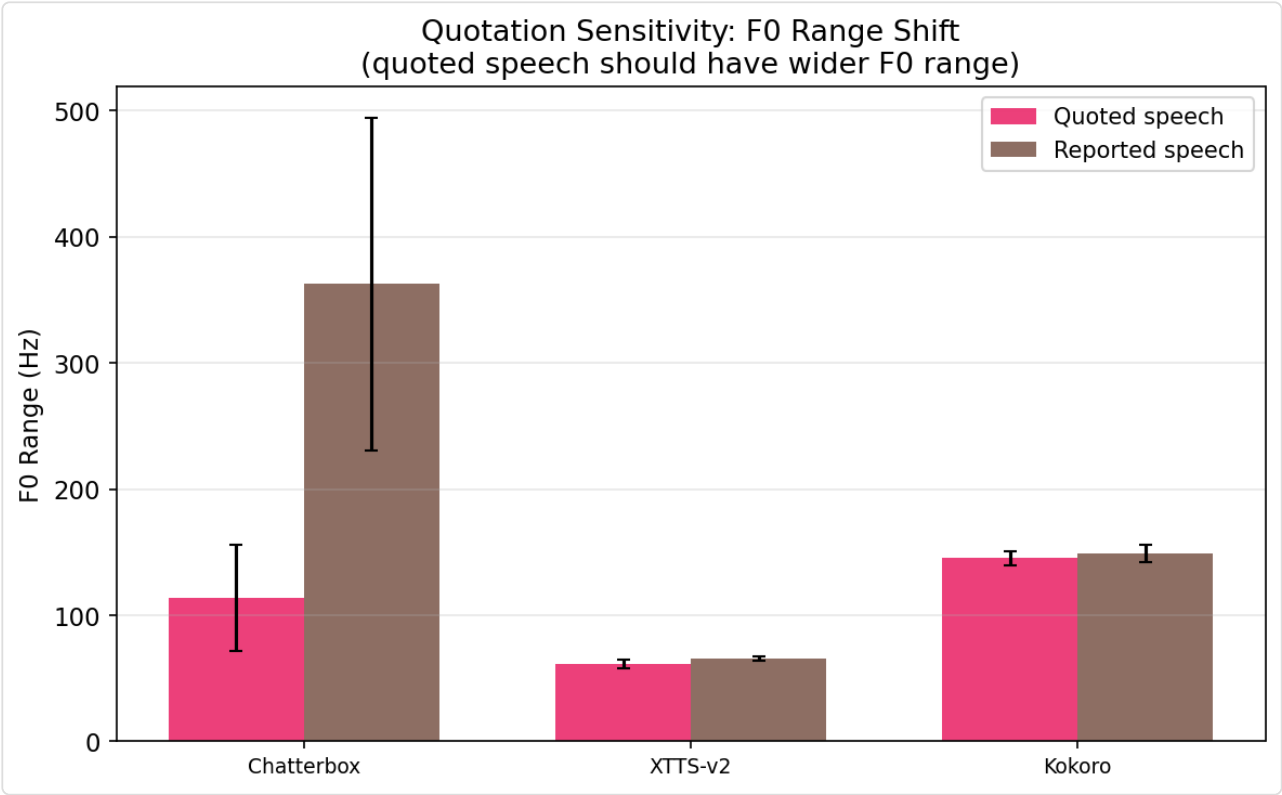


Model	Ellipsis F0 (Hz/s)	Period F0 (Hz/s)	Ellipsis Pause (ms)	Period Pause (ms)	F0 Diff (Hz/s)	Co
Chatterbox	-124.6	-148.8	235.0	220.0	24.2	0.3 (sn
XTTS-v2	-60.9	-43.8	590.0	690.0	-17.1	-18 (la
Kokoro (no-adaptation baseline)	63.1	2.4	500.0	505.0	60.7	1.5 (la

Kokoro shows the largest F0 difference between ellipsis and period (+60.7 Hz/s, large d). Chatterbox also differentiates (+24.2 Hz/s). XTTS-v2 shows negligible difference (-17.1 Hz/s, small d). All comparisons are based on n=2 per condition per model.

7 Quotation Sensitivity

F0 range and mean shift between quoted and reported speech. Shift flag = F0 range shift > configured threshold.



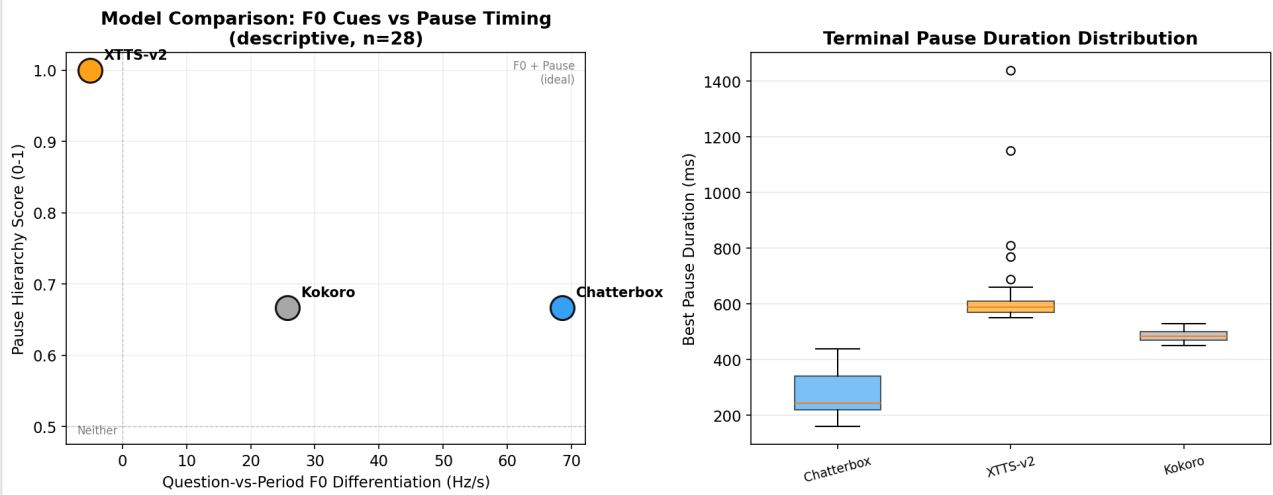
Model	F0 Mean Shift (Hz)	F0 Range Shift (Hz)	Shift Detected	Cohen's d (range)
Chatterbox	3.1	249.0	YES	-1.3 (large)
XTTS-v2	1.8	3.9	NO	-0.7 (medium)
Kokoro (no-adaptation baseline)	5.8	3.7	NO	-0.3 (small)

Chatterbox shows a quotation F0 range shift flagged by the threshold (n=2 per condition). XTTS-v2 and Kokoro do not. The threshold (>10 Hz range shift) is arbitrary; with n=2, this is anecdotal evidence at best.

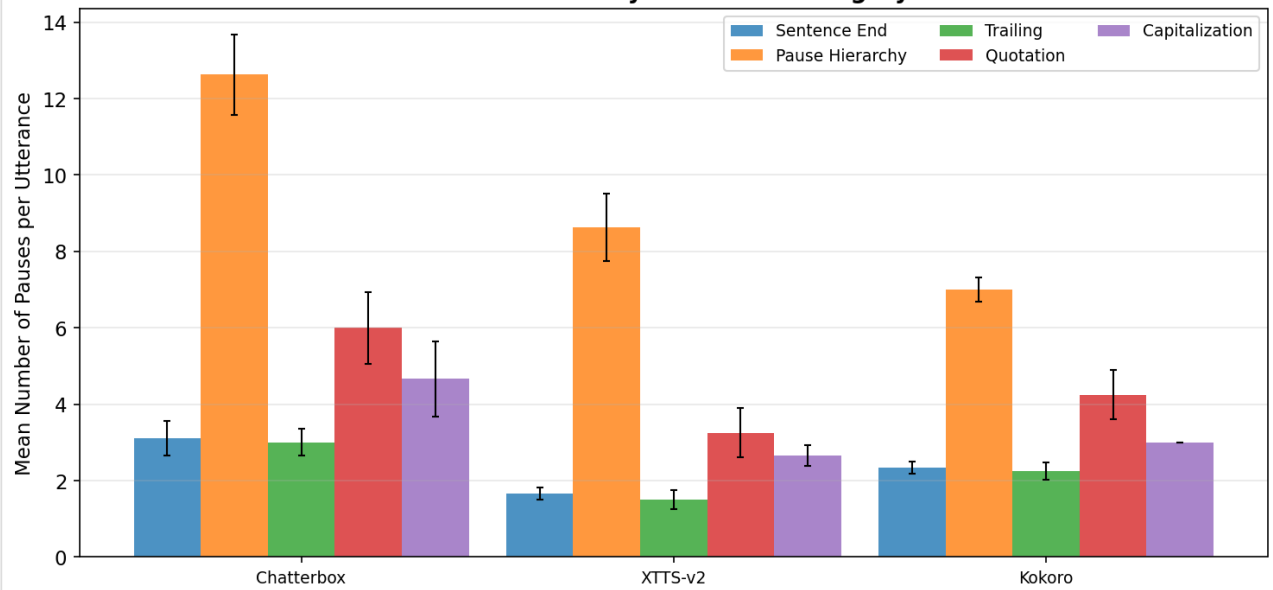
8 Summary

The scatter plot shows the relationship between F0-based and pause-based punctuation sensitivity.

Punctuation Sensitivity: Summary



Pause Count by Utterance Category



Model	Question Rise (Hz/s)	Pause Gradient (ms)	Pause Hierarchy
Chatterbox	68.5	5.6	0.67
XTTS-v2	-5.0	101.5	1.0
Kokoro (no-adaptation baseline)	25.7	30.8	0.67

Descriptive observation. F0 sensitivity and pause ordering appear to vary independently across these 3 models in this 28-item probe. Chatterbox shows stronger F0 differentiation; XTTS-v2 shows stronger pause ordering. These are hypothesis-generating observations, not validated findings.

9 Capitalisation Sensitivity

WARNING: 1 item per condition. These are anecdotes, not evidence.

Model	RMS Boost in ALL CAPS	F0 Boost in ALL CAPS
Chatterbox	True	False
XTTS-v2	True	0.0
Kokoro (no-adaptation baseline)	True	N/A

All models increase RMS amplitude for ALL CAPS. F0 boost is inconsistent. With $n=1$ per condition, no conclusions should be drawn.

10 Methodological Limitations

- **Sample size.** 28 utterances per model with 2-6 items per subcategory. Insufficient for statistical inference or model ranking.
- **No perceptual ground truth.** All metrics are acoustic proxies (VAD pauses, pyin F0). No listening tests, MOS, ABX, or preference judgments were collected. These metrics have unknown correlation with human perception of prosody.
- **No forced alignment.** Pauses are pooled across all internal RMS-detectable silences, not anchored to specific punctuation tokens. A silence anywhere in the utterance before 95% duration counts as an "internal pause" regardless of where punctuation occurs.
- **Crude VAD.** Energy threshold ($0.05 \times \text{max RMS}$) is fragile across models with different loudness, noise floor, vocoder artifacts, and trailing silence policies.
- **F0 extraction noise.** librosa.pyin can fail on creaky voice, unvoiced endings, and vocoder artifacts in short generated utterances. README and code acknowledge this.
- **Arbitrary thresholds.** Gate (150 ms), quotation shift (>10 Hz F0 range), and internal pause cutoff (95% duration) are all heuristic values without empirical justification.
- **No text normalization audit.** TTS frontends may normalize punctuation before synthesis. This is not inspected or logged in the current pipeline.
- **Unfair model comparison.** Kokoro lacks voice cloning and uses a different speaker, prosody prior, and conditioning mode. It serves as a "no-adaptation baseline" but should not be compared directly in rankings.
- **Capitalization test conflates factors.** "STOP!" vs "Stop!" vs "stop!" differs in capitalization, token frequency, and potentially phoneme-level priors. $n=1$.

- **No punctuation-stripped controls.** Without a baseline condition where punctuation is removed, we cannot isolate whether models respond to the punctuation mark itself or to tokenizer-side effects of the surrounding text.

Generated from the Punctuation Sensitivity Probe — descriptive analysis of 84 utterances across Chatterbox, XTTS-v2, and Kokoro.

Report generated with Python 3.12 • stdlib minimal dependencies.